

AI-Powered Smart Traffic Management System

Abstract

Traffic congestion remains one of the most significant challenges in modern urban environments, leading to increased travel times, fuel consumption, and environmental pollution. Traditional traffic signal systems typically operate using fixed timing schedules that cannot adapt effectively to changing traffic conditions. This project presents an AI-powered Smart Traffic Management System that utilizes Computer Vision and Machine Learning techniques to monitor traffic flow and dynamically optimize signal timing.

The proposed system employs YOLOv8s for vehicle detection and ByteTrack for multi-object tracking. Vehicle counts, traffic flow, and queue information are analyzed in real time to determine the most efficient traffic signal phase. The system also incorporates emergency vehicle prioritization to improve response times for critical services. Experimental results demonstrate strong detection performance with a mAP@50 of 92%, mAP@50-95 of 77.4%, precision of 88.7%, and recall of 85%.

1. Introduction

Urban traffic congestion has become a growing concern due to increasing vehicle ownership and population density. Conventional traffic management systems often rely on static timing mechanisms that cannot respond to real-time traffic demands, resulting in inefficient traffic flow and increased delays.

Recent advances in Artificial Intelligence and Computer Vision provide opportunities for intelligent traffic management systems capable of adapting to changing road conditions. This project aims to develop a smart traffic control solution that automatically analyzes traffic conditions and adjusts traffic signals accordingly.

The primary objectives of this project are:

- Detect and classify vehicles in real time.
- Track vehicles across video frames.
- Estimate traffic density and vehicle flow.
- Dynamically adjust traffic signal timing.
- Prioritize emergency vehicles when detected.
- Improve overall traffic efficiency.

2. System Overview

The proposed system consists of several integrated modules that work together to monitor and manage traffic conditions.

The workflow of the system is as follows:

Camera Input



YOLOv8 Vehicle Detection



ByteTrack Vehicle Tracking



Traffic Analysis



Adaptive Traffic Controller



Traffic Signal Decision

Video streams captured from traffic cameras are processed by the detection module. Detected vehicles are then tracked across frames to avoid duplicate counting and to obtain traffic statistics. The adaptive controller analyzes traffic conditions and determines the optimal signal timing based on current demand.

3. Dataset Preparation

One of the major challenges encountered during development was the lack of a single high-quality dataset containing all required vehicle classes in sufficient quantities.

To address this issue, a custom dataset was constructed by combining multiple datasets collected from Roboflow. Each vehicle category was sourced from specialized datasets and merged into a unified training dataset.

The final dataset contains approximately 30,000 annotated images covering six vehicle classes:

- Car
- Truck
- Bus
- Motorcycle
- Ambulance
- Fire Truck

This custom dataset provided a diverse collection of images under various lighting conditions, viewing angles, and traffic scenarios, improving model robustness and generalization.

4. Model Development

The vehicle detection model was developed using YOLOv8s due to its balance between accuracy and computational efficiency.

Training configuration:

- Model: YOLOv8s
- Epochs: 50
- Batch Size: 60
- Image Size: 640 × 640
- Optimizer: Adam

The model was trained on the custom vehicle dataset and optimized to accurately detect both common and emergency vehicles.

5. Vehicle Tracking

Vehicle tracking was implemented using ByteTrack, a state-of-the-art multi-object tracking algorithm.

ByteTrack assigns a unique identifier to each detected vehicle and maintains tracking consistency across video frames. This enables:

- Accurate vehicle counting.

- Vehicle flow measurement.
- Prevention of duplicate counts.
- Queue length estimation.

The tracking module serves as a critical component in generating reliable traffic statistics for decision-making.

6. Adaptive Traffic Control System

The adaptive controller dynamically determines traffic signal durations based on real-time traffic conditions.

The controller analyzes several traffic indicators, including:

- Vehicle flow.
- Queue length.
- Road demand.
- Signal starvation prevention.

Using these metrics, the system allocates green-light durations according to current traffic requirements rather than relying on fixed timing schedules.

Additionally, the system includes emergency vehicle prioritization. When an ambulance or fire truck is detected, the controller can prioritize the corresponding traffic direction to facilitate faster emergency response.

7. Results and Evaluation

The trained model achieved strong performance across all evaluation metrics.

Performance Results:

- mAP@50: 92%
- mAP@50-95: 77.4%
- Precision: 88.7%
- Recall: 85%

These results indicate that the model is capable of accurately detecting and classifying vehicles under various traffic conditions.

The integration of YOLOv8s with ByteTrack provided reliable real-time monitoring capabilities while maintaining efficient system performance.

8. Challenges Faced

The most significant challenge during development was dataset preparation.

No single publicly available dataset contained all required vehicle classes with sufficient image quantity and quality. As a result, multiple datasets had to be collected, cleaned, balanced, and combined into a unified dataset.

This process required substantial effort to ensure consistency between annotations, image quality, and class representation.

Despite these challenges, the final dataset successfully supported the training of a robust vehicle detection model capable of handling both regular and emergency vehicles.

9. Conclusion

This project demonstrates the successful development of an AI-powered Smart Traffic Management System that combines Computer Vision and Machine Learning techniques to improve traffic control efficiency.

By integrating YOLOv8s for vehicle detection, ByteTrack for tracking, and an adaptive signal controller, the system is capable of monitoring traffic conditions in real time and making intelligent traffic management decisions.

The achieved detection performance and system functionality highlight the potential of AI-driven solutions in addressing modern traffic challenges and supporting future smart city initiatives.